INCH-POUND
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SUPERSEDING
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07 May 2002

PERFORMANCE SPECIFICATION

HOSE AND HOSE ASSEMBLIES, NONMETALLIC: ELASTOMERIC, LIQUID FUEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standard collapsible and non-collapsible, lightweight collapsible, and low temperature non-collapsible elastomeric, liquid fuel transfer hose and hose assemblies.

1.2 Classification. The hose and hose assemblies are of the following types, sizes, classes, styles, material, and lengths (see 6.2).

1.2.1 Type. The types of hose and hose assemblies are as follows:

Type A - Collapsible (see 6.5.1)
Type B - Non-collapsible (see 6.5.2)
Type C - Collapsible, lightweight (2-, 3-, 4-, and 6-inch nominal diameter only)
Type D - Non-collapsible low temperature

Comments, suggestions, or questions on this document should be addressed to U.S. Army Tank-Automotive Research, Development and Engineering Center, ATTN: RDTA-SIE-ES-SI MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or sent by email to mailto: usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.2 Size. The sizes of the hose and hose assemblies, as determined by the nominal inside diameter of the hose, are as follows:

Size 01	–	0.50 inches	Size 06	–	2.00 inches
Size 02	–	0.75 inches	Size 07	–	2.50 inches
Size 03	–	1.00 inches	Size 08	–	3.00 inches
Size 04	–	1.25 inches	Size 09	–	4.00 inches
Size 05	–	1.50 inches	Size 10	–	6.00 inches

1.2.3 Class. The classes of hose and hose assemblies will be as follows:

Class 0	-	Without fittings
Class 1	-	Male fitting on one end and female fitting on other end
Class 2	-	Male fitting both ends
Class 3	-	Female fitting both ends
Class 4	-	Sexless both ends

1.2.4 Style. The end fitting style will be as follows:

Style O	-	Bulk hose
Style A	-	Cam-locking end fittings
Style B	-	Re-attachable (reusable) compression end fitting up to 4-inch
Style C	-	Grooved end fittings
Style D	-	One-time-use internal expanded end fittings up to 4-inch
Style E	-	Quick-disconnect, sexless couplings (see 3.3.5.5)
Style F	-	Quick-disconnect, 2-inch sexless coupling on one hose end and 3-inch sexless coupling on the other hose end (see 3.3.5.5)

1.2.5 Material. The material used in the end fittings will be as follows:

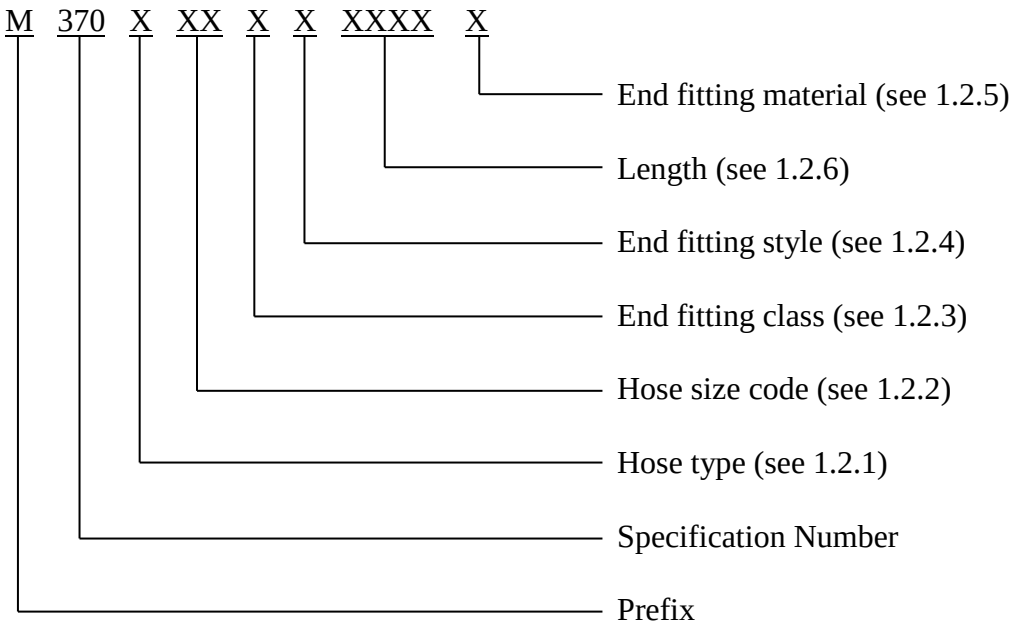
Material A	-	Aluminum
Material B	-	Brass
Material C	-	Corrosion resistant steel
Material D	-	Steel (carbon)
Material O	-	Other
Material S	-	Contractor Specified

1.2.6 Length. The hose and hose assembly length will be expressed in feet, to four significant figures, with the last digit indicating tenths of a foot. The following are examples of the length code:

5000	-	500 feet 0 inch
0505	-	50 feet 6 inches

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1.3 Part or identifying number. This part or identifying number (PIN) of hose and hose assemblies covered by this specification is created in the following form (see 6.2).



Example: M370B101A5000C (A reinforced, non-collapsing hose conforming to this specification, 6-in. in diameter, style of male one end female other end, cam-locking class fittings, a length of 500 ft, and fittings made of aluminum).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NORTH ATLANTIC TREATY ORGANIZATION (NATO)

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AFLP 3747	-	Guide Specifications (Minimum Quality Standards) For Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40 AND F-44)

COMMERCIAL ITEM DESCRIPTIONS

A-A-52592	-	Pipe Fittings: One Or More Ends Grooved
A-A-59326	-	Coupling Halves, Quick-Disconnect, Cam-Locking Type
A-A-59326/10	-	Coupling Half, Cap, Dust, Type IX
A-A-59326/11	-	Coupling Half, Plug, Dust, Type X
A-A-59377	-	Coupling Assembly, Quick-Disconnect, Sexless Type

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-83133	-	Turbine Fuels, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35 and JP-8 +100 (NATO F-37)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ARMY REGULATIONS

AR 70-75	-	Survivability of Army Personnel and Material

(Copies of these documents are available online at http://www.apd.army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	-	Pipe Threads, General Purpose (Inch)

(Copies of this document are available online from http://www.asme.org/.)

ASTM INTERNATIONAL

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ASTM D380	-	Standard Test Methods for Rubber Hose
ASTM D381	-	Standard Test Method for Gum Content in Fuels by Jet Evaporation
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D413	-	Standard Test Methods for Rubber Property - Adhesion to Flexible Substrate
ASTM D471	-	Standard Test Method for Rubber Property - Effect of Liquids
ASTM D573	-	Standard Test Method for Rubber – Deterioration in an Air Oven
ASTM D746	-	Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact
ASTM D790	-	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D975	-	Standard Specification for Diesel Fuel Oils
ASTM D1053	-	Standard Test Methods for Rubber Property - Stiffening at Low Temperatures: Flexible Polymers and Coated Fabrics
ASTM D1149	-	Standard Test Method for Rubber Deterioration – Cracking in an Ozone Controlled Environment
ASTM D1655	-	Standard Specification for Aviation Turbine Fuels
ASTM D2137	-	Standard Test Methods for Rubber Property - Brittleness Point of Flexible Polymers and Coated Fabrics
ASTM D3389	-	Standard Test Method for Coated Fabrics Abrasion Resistance (Rotary Platform Abrader)
ASTM D4814	-	Standard Specification for Automotive Spark-Ignition Engine Fuel

(Copies of these documents are available online from http://www.astm.org.)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C606	-	Grooved and Shouldered Joints

(Copies of these documents are available online from http://www.awwa.org/.)

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN)

EN 228	-	Automotive Fuels – Unleaded Petrol – Requirements and Test Methods
EN 590	-	Automotive Fuels – Diesel – Requirements and Test Methods

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(Copies of these documents are available online from https://www.cen.eu/.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS-STD-595	-	Colors Used in Government Procurement

(Copies of these documents are available online from http://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), hose/hose assembly shall be subjected to first article inspections in accordance with 4.1.1.

3.2 Materials. The manufacturer shall select materials capable of meeting all the operational and environmental requirements specified herein. The hose assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide against corrosion and deterioration during storage and operational conditions experienced.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recovered materials (see 6.5.3) shall be used to the maximum extent practicable, provided the components meet all other requirements of this specification. Used, rebuilt, or remanufactured components shall not be incorporated into the hose or hose assemblies.

3.3 Design and construction. The hose shall be finished lengths, as specified, consisting of a tube (see 6.5.4), reinforcement (see 6.5.5), stiffening (non-collapsible only), and a cover (see 6.5.6). Hose assemblies shall be the defined types, sizes, and lengths of hose, equipped with end fittings as specified (see 4.3).

3.3.1 Hose dimensions.

3.3.1.1 Type A, B and D hose size. All type A, B, and D hoses shall conform to the size requirements as specified in table I (see 4.3.1.1).

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TABLE I. Type A, B, and D hose diameters and tolerances.

Size code	Inside diameter (in.)	Outside diameter (in.)
01	0.50 ±0.03	1.06 ±0.03
02	0.75 ±0.03	1.37 ±0.03
03	1.00 ±0.06	1.62 ±0.06
04	1.25 ±0.06	1.87 ±0.06
05	1.50 ±0.06	2.12 ±0.06
06	2.00 ±0.06	2.66 ±0.06
07	2.50 ±0.06	3.03 ±0.06
08	3.00 ±0.06	3.75 ±0.06
09	4.00 ±0.06	4.75 ±0.06
10	6.00 ±0.06	6.87 +0.12/-0.06

3.3.1.2 Type C hose size and weight. Type C hoses shall conform to the size and weight requirements in table II (see 4.3.1.2).

TABLE II. Type C hose diameter, tolerance, and weight.

Size code	Inside diameter (in.)	Weight (lb/ft)
06	2.06 ±0.06	0.5 maximum
07	2.56 ±0.06	0.7 maximum
08	3.06 ±0.06	0.75 maximum
09	4.06 ±0.06	1.0 maximum
10	6.06 ±0.06	1.5 maximum

3.3.2 Length. The length of hose and hose assembly shall be the actual hose length exclusive of the end fittings. The tolerance on the length shall be ±2 percent (see 4.3.2).

3.3.3 Tube and cover. The tube shall be of petroleum-resistant compounds (see 3.4.2) and shall be smooth, free of defects (e.g., holes, pitting, blisters, etc.), and of a uniform thickness. The cover shall be of petroleum-resistant compounds (see 3.4.2) and shall be free of defects (e.g., holes, pitting, blisters, etc.) and be ultraviolet-stable and ozone-resistant. The tube and cover shall meet all the performance requirements as specified herein (see 4.3.3).

3.3.4 Static bond. Unless otherwise specified (see 6.2), the hose shall contain a continuous embedded static wire(s) or other electrical conductor(s). The conductor(s) shall provide a continuous electrical bond between the hose ends to include fittings. The maximum resistance shall not be greater than 1.5 ohms per linear foot of hose. The bond shall not break when the hose is coiled, sharply bent, or subjected to internal or external pressures as specified (see 4.3.4 and 4.3.6.7).

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3.3.5 End fittings. End fittings shall be as identified by class, style, and material and sized for the hose size specified. Non-corrosive devices (see 6.4) shall be used for the attachment of end fittings to the hose. All end fittings shall be securely attached to the hose to prevent slippage or leakage under normal operation and when subjected to hydrostatic proof pressure. Unless otherwise specified (see 6.2), the end fittings shall be provided with protective, closed-end, reusable dust plugs or caps. Means (i.e. chain) shall be used to captively retain the dust plugs or caps to the end fittings (see 4.3.5).

3.3.5.1 Coupling halves, quick-disconnect cam-locking type. Cam locking end fittings shall be in accordance with A-A-59326. Unless otherwise specified, dust caps or plugs for cam-locking end fittings shall be in accordance with A-A-59326/10 and A-A-59326/11, respectively (see 6.2). The coupling halves and gaskets shall be compatible for use with the fuels listed in 3.4.2

3.3.5.2 Grooved end fittings. Grooved end of the hose fitting adapters shall be in accordance with the Cut Grooving Dimensions – Steel, Aluminum, Brass, and Other Metallic Pipe of IPS Dimensions table of AWWA C606. Hose adapter section of the grooved end fitting shall be sized to fit the inside diameter of hose as specified in table I for type A, B, and D hoses and table II for type C hoses. The grooved end fittings shall be rated at 150 psig.

3.3.5.3 Re-attachable screw-on hose fittings. The re-attachable compression type fitting (see figure 1) shall consist of an outer sleeve, inner body/tailpiece, and wire compression grip. The tailpiece shall have one end threaded with American National Standard Taper Pipe threads (NPT) in accordance with ASME B1.20.1, for attachment to existing assemblies and a hose shank type end for hose attachment. The hose shank end of the tailpiece shall have the sleeve threaded over the tailpiece that allows the hose to be compressed and firmly secured to the tailpiece. When the sleeve is fully engaged it shall extend beyond the hose end of the tailpiece but shall not cover any of the NPT thread. The sleeve shall have at least four flat surfaces to permit the use of a standard wrench for tightening or loosening.

3.3.5.3.1 Pull strength (assemblies with re-attachable fittings). The couplings, when installed in accordance to the manufacturers' instructions, shall withstand a repeated, longitudinal pull of 450 pounds without the fittings becoming detached from the hose (see 4.3.5.1).

3.3.5.4 One-time-use end fittings. The one-time-use internal expanded fittings shall have an internal expanding tailpiece and external ferrule. The tailpiece shall grip and press the hose wall against the external ferrule. When installed, these fittings shall have an inside diameter conforming to the nominal inside diameter of the hose.

3.3.5.5 Quick-disconnect, sexless couplings. The sexless couplings shall be class A in accordance with A-A-59377. The couplings shall be reusable.

3.3.6 Physical and chemical requirements of bulk hose. The requirements for the physical and chemical properties, by hose type, shall be as specified in table III. The requirement column of the table shall be the minimum level for acceptability. Test procedures as

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listed in the ASTM column shall be used as augmented or modified by the noted test paragraphs (see 4.3.6).

TABLE III. Hose physical and chemical properties.

Physical properties	Hose type				Requirements	Test method	
	A	B	C	D		ASTM	Test
Tensile strength, original:							
Tube	X	X	X	X	1250 psi, minimum	D412	4.3.6.1
Cover	X	X		X	1250 psi, minimum		
			X		1000 psi, minimum		
Tensile strength, after immersion in:							
Fuel	X	X	X	X	Minimum of 40% of original, or 600 psi	D471	4.3.6.2
Water (for 14 days)			X		Minimum of 40% of original, or 600 psi	D471	4.3.6.2.1
Water (for 42 days)			X		Minimum of 25% of original, or 300 psi		
Ultimate elongation, original	X	X		X	200%, minimum	D412	4.3.6.1
			X		400 % minimum		
Ultimate elongation, after immersion in:							
Fuel	X	X		X	100%, minimum	D471	4.3.6.2
			X		200%, minimum		
Water (for 14 days)			X		40% of original, minimum	D471	4.3.6.2.1
Water (for 42 days)			X		25% of original, minimum		
Volume increase, after immersion:							
Tube	X	X		X	60%, maximum	D471	4.3.6.2
			X		40%, maximum		
Cover	X	X		X	100%, maximum		
			X		70%, maximum		

TABLE III. Hose physical and chemical properties - Continued.

Physical properties	Hose type				Requirements	Test method	
	A	B	C	D		ASTM	Test
Adhesion between reinforcement and:							
Tube	X	X	X	X	10 lbf/in[1], minimum	D413	4.3.6.3.1
Cover	X	X	X	X	10 lbf/in[1], minimum		
Adhesion after immersion in fuel:							

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Tube/reinforcement	X	X		X	6 lbf/in[1], minimum	D413	4.3.6.3.2
Cover/reinforcement	X	X		X	6 lbf/in[1], minimum		
Adhesion after fill test (fuel):							
Tube			X		8 lbf/in[1], minimum	D413	4.3.6.3.3
Cover			X		9 lbf/in[1], minimum		
Physical properties	Hose type				Requirements	Test method	
	A	B	C	D	A	B	C
Low temperature flexibility:							
Young's modulus	X	X			10,000 psi, maximum	D1053	4.3.6.4.1
Relative torsional modulus			X		5, minimum		
Brittleness				X	No evidence of cracking	D471 D746 D2137	4.3.6.4.2
Modulus of elasticity tube				X	10,000 psi, maximum	D790	4.3.6.4.3
Modulus of elasticity cover				X	5000 psi, maximum		
Resistance to aging:							
Tensile strength	X	X	X	X	70% of original, minimum	D 573	4.3.6.5
Ultimate elongation	X	X		X	50% of original, minimum		
			X		70% of original, minimum		
Existent gum	X	X		X	20 mg/100 mL of test fluid, maximum	D381 D471	4.3.6.6
			X		6 mg/100 mL of test fluid, maximum		

TABLE III. Hose physical and chemical properties - Continued.

Physical properties	Hose type				Requirements	Test method	
	A	B	C	D		ASTM	Test
Crush resistance:							
Under load -		X		X	85% of original diameter, minimum	---	4.3.6.7
After load release -		X		X	95% of original diameter, minimum		
Bend radius		X		X	No deformation, crimping, or loss of pressure	---	4.3.6.8
Ozone resistance	X	X	X	X	No visible cracking (viewed under 7x lens)	D518 D1149	4.3.6.9
Abrasion resistance (when required – see 6.2)			X		Report number of cycle needed to expose reinforcing material.	D3389	4.3.6.10

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NOTES:
1. lbf/in. = Pounds-force per inch

3.3.7 Hose assemblies.

3.3.7.1 Proof pressure. Hose assemblies shall not leak nor have deleterious imperfections when subjected to the appropriate proof pressure, with respect to hose size, as specified in table IV (for hose assemblies constructed with type A, B, or D hose) or table V (for hose assemblies constructed with type C hose). There shall be no slippage or pullout of the fittings from the hose. For Type A, B, and D hose, the length of the hose shall not change more than ±7% when subjected to proof pressure. For type C hose, the length of hose shall not change more than ±2% when subjected to proof pressure. Type A and C hose shall not twist more than one-half turn (180°) in 25 ft when subjected to proof pressure (see 4.3.7.1).

TABLE IV. Pressure requirements for type A, B, and D hose.

Hose Nominal I.D. (in)	Proof pressure, minimum (psig)	Burst pressure, minimum (psig)	Working pressure, maximum (psig)[1]
0.50	250	500	125
0.75	250	500	125
1.00	250	500	125
1.25	250	500	125
1.50	250	500	125
2.00	200	400	100
2.50	200	400	100
3.00	150	300	75
4.00	150	300	75
6.00	150	300	75

NOTES:
1. Recommended.

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TABLE V. Pressure requirements for type C hose.

Hose Nominal I.D. (in)	Proof pressure, minimum (psig)	Burst pressure, minimum (psig)	Working pressure, maximum (psig)[1]
2.00	225	600	150
2.50	225	600	150
3.00	225	600	150
4.00	225	600	150
6.00	225	600	150

NOTES:
1. Recommended.

3.3.7.2 Burst pressure. The hoses assemblies shall have a minimum burst pressure as specified in table IV (for hose assemblies constructed with type A, B, or D hose) or table V (for hose assemblies constructed with type C hose). For hose assemblies constructed with type C hoses, the hoses shall not fail radially (see 4.3.7.2).

3.4 Interface. For verification of interface, see 4.4.

3.4.1 Connection to existing equipment. The hose or hose assemblies defined herein are required to interface with existing equipment and hose assemblies. The interface points, end fittings, shall be controlled. Re-attachable end fittings shall be as specified herein; A-A-59326 shall be used for cam-locking fittings; A-A-52592 shall be used for grooved end fittings; and A-A-59377 shall be used for quick disconnect sexless couplings.

3.4.2 Fuel and water compatibility. Finished hoses shall be resistant to, have no deleterious effects upon, and not be degraded by water and the following kerosene-based petroleum fuels. Meeting the hose physical and chemical properties requirements (see 3.3.6) with the specified test fluid shall provide sufficient evidence that the finished hoses conform to the fuel compatibility requirement (see 3.3.3).

a. AFLP 3747 (NATO F-24)

b. ASTM D975 (Diesel-US commercial)

c. ASTM D1655 (Jet A-1)

d. ASTM D4814.

e. EN 228.

f. EN 590.

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g. MIL-DTL-5624 (JP-5) (NATO F-44) (see 6.7)

h. MIL-DTL-83133 (JP-8) (NATO F-34) (see 6.7)

i. MIL-DTL-83133 (NATO F-35)

3.5 Ownership and support. For verification of ownership and support, see 4.5.

3.5.1 Safety. Hose and hose assemblies shall comply with all U.S. HAZMAT, OSHA and safety requirements regarding the prevention of inadvertent fuel discharge or leakage during operation, storage and maintenance.

3.5.2 Human factors engineering. Hose and hose assemblies shall be capable of being employed by the 5th percentile female to the 95th percentile male military personnel wearing gloves or arctic mittens, or equipped with Mission Orientated Protective Posture (MOPP) IV gear (see 6.8) without special tools or support equipment.

3.5.3 Cover color. Unless otherwise specified, the cover color shall be in accordance with SAE AMS-STD-595 color ID 33446 or in accordance with SAE AMS-STD-595 color ID 37038 (see 6.2).

3.5.4 Hose marking. The letters shall be a minimum of 0.18 in. high. Hoses shall be marked at 10-ft intervals or by continuous impression branding, provided that the pattern shall be repeated every 5 ft. or less. Marking color shall contrast with the hose cover color. Each hose shall be marked with a longitudinal stripe of a contrasting color, not less than 0.125 in. wide, and extending the full length of the hoseline. The marking may be within the longitudinal stripe, provided the stripe is at least 0.2 in. greater than the height of the letters. The marking may substitute for the longitudinal stripe, provided it is continuous and extends the full length of the hoseline. Labeling and longitudinal stripe shall remain clear, distinct and legible for the life of the hoseline. Handling, bending, water, oil, and environmental effects shall not adversely affect the marking or stripe. Hose and hose assemblies shall be permanently marked with the following information unless otherwise specified (see 6.2):

a. Date of manufacture (quarter and year)

b. Manufacturer’s name or trademark

c. “MIL-PRF-370”

d. Hose type (Example: “TYPE C”)

e. “LIQUID FUEL”

3.5.5 Chemical, biological, radiological, and nuclear (CBRN). Hose and hose assemblies shall be CBRN contamination survivable and capable of operating in a CBRN

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environment. Nuclear survivability is not required. The hose and hose assemblies shall be capable of being operated and serviced by personnel wearing Mission Orientated Protective Posture (MOPP IV) chemical, biological and radiological clothing (see 6.8) without special tools or support equipment. The design of the hose and hose assemblies shall minimize the collection and retention of contaminants and decontaminates. The hose and hose assemblies shall be decontaminable to negligible risk levels as defined in Army-approved NBC contamination survivability criteria in accordance with AR 70-75 (see 4.5.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article test (FAT) (see 4.1.1).

b. Conformance inspection (CI) (see 4.1.2).

4.1.1 FAT. When specified (see 6.2), the contractor shall perform FAT and inspections on production hose and hose assemblies in accordance with table VI and paragraphs 4.4 through 4.5 of this specification.

TABLE VI. FAT application and test methods.

Test	Hose type				Test description	Requirement Paragraph	Test Paragraph
	A	B	C	D			
101			X		Weight	3.3.1.2	4.3.1.2
102	X	X	X	X	Static bond (at pressure)	3.3.4	4.3.4
103	X				Pull strength with re-attachable fittings	3.3.5.3.1	4.3.5.1
104	X	X	X	X	Tensile strength, original (tube and cover)	3.3.6, Table III	4.3.6.1
105	X	X	X		Tensile strength after fuel immersion	3.3.6, Table III	4.3.6.2

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106			X		Tensile strength after water immersion (14 days)	3.3.6, Table III	4.3.6.2.1
107			X		Tensile strength after water immersion (42 days)	3.3.6, Table III	4.3.6.2.1
108	X	X		X	Ultimate elongation, original	3.3.6, Table III	4.3.6.1
109	X	X	X	X	Ultimate elongation after fuel immersion	3.3.6, Table III	4.3.6.2
110			X		Ultimate elongation after water immersion (14 days)	3.3.6, Table III	4.3.6.2.1
111			X		Ultimate elongation after water immersion (42 days)	3.3.6, Table III	4.3.6.2.1
112	X	X	X	X	Volume increase after fuel immersion (tube and cover)	3.3.6, Table III	4.3.6.2
113	X	X	X	X	Adhesion (between reinforcement and tube), original	3.3.6, Table III	4.3.6.3.1
114	X	X	X	X	Adhesion (between reinforcement and cover), original	3.3.6, Table III	4.3.6.3.1
115	X	X		X	Adhesion (tube/reinforcement), after immersion in fuel	3.3.6, Table III	4.3.6.3.2
116	X	X		X	Adhesion (cover/reinforcement), after immersion in fuel	3.3.6, Table III	4.3.6.3.2
117			X		Adhesion (tube/reinforcement), after fuel fill test	3.3.6, Table III	4.3.6.3.3
118			X		Adhesion (cover/reinforcement), after fuel fill test	3.3.6, Table III	4.3.6.3.3
119	X	X			Low temperature flexibility	3.3.6, Table III	4.3.6.4.1
120			X		Torsional rotation	3.3.6, Table III	4.3.6.4.1

TABLE VI. FAT application and test methods – Continued.

Test	Hose type				Test description	Requirement Paragraph	Test Paragraph
	A	B	C	D			
121				X	Brittleness	3.3.6, Table III	4.3.6.4.2
122				X	Modulus of elasticity (tube and cover)	3.3.6, Table III	4.3.6.4.3
123	X	X	X	X	Resistance to aging	3.3.6, Table III	4.3.6.5
125		X			Crush resistance (under load and after load release)	3.3.6, Table III	4.3.6.7

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126		X		X	Bend radius	3.3.6, Table III	4.3.6.8
127	X	X	X	X	Ozone resistance	3.3.6, Table III	4.3.6.9
128			X		Abrasion resistance	3.3.6, Table III	4.3.6.10
129	X	X	X	X	Proof pressure	3.3.7.1	4.3.7.1
130	X		X		Twist (at pressure)	3.3.7.1	4.3.7.1
131	X	X	X	X	Burst pressure	3.3.7.2	4.3.7.2

4.1.2 Conformance inspection (CI). When specified (see 6.2), CI shall be performed on all hose and hose assembly production units. CI shall include the examinations of table VII and the test as specified in 4.3.7.1. Nonconformance to any specified requirement of this specification and any additional requirements of the contract, purchase order, or scope of work (SOW) shall be cause for rejection.

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TABLE VII. Examination schedule.

Examination	Examination description	Method of inspection
101	Materials not as specified (see 3.2).	Visual
102	Hose type not as specified (see 3.3).	Visual/Doc.[1]
103	Hose size not as specified (see 3.3.1).	Visual/Doc./SIE[2]
104	Hose length not as specified (see 3.3.2).	SIE
105	Tube not as specified (see 3.3.3).	Visual/Doc.
106	Cover not as specified (see 3.3.3).	Visual/Doc.
107	End fittings not as specified (see 3.3.5).	Visual/Doc.
108	Dust plug(s) or cap(s) not as specified (see 3.3.5).	Visual/Doc.
109	Cam-locking end fitting not as specified (see 3.3.5.1).	Visual/Doc.
110	Grooved end fitting not as specified (see 3.3.5.2).	Visual/Doc.
111	Re-attachable screw-on hose coupling not as specified (see 3.3.5.3).	Visual/Doc.
112	One-time-use end fitting not as specified (see 3.3.5.4).	Visual/Doc.
113	Quick-disconnect, sexless coupling not as specified (see 3.3.5.5).	Visual/Doc.
114	Physical/chemical requirements not as specified (see 3.3.6).	Visual/Doc.
115	Hose assemblies not as specified (see 3.3.7).	Visual/Doc.
116	Interface requirements not as specified (see 3.4).	Visual/Doc.
117	Finished hose not resistant to fuels as specified (see 3.4.2).	Doc.
118	Ownership/support requirements not as specified (see 3.5).	Visual/Doc.
119	Cover color not as specified (see 3.5.3).	Visual/Doc.
120	Marking not as specified (see 3.5.4).	Visual/Doc.
121	Packaging not as specified (see 5 and contract/SOW).	Visual/Doc.

NOTES:
1. Doc. = Documentation
2. SIE = Standard inspection equipment

4.2 Test conditions, schedules, other required reports or certifications, and prohibitions.

4.2.1 Test conditions. Unless otherwise specified, all testing and examinations shall be conducted at 77°F ± 3°F or, with prior Government permission, all values obtained using ambient temperature and humidity must be corrected to this required temperature in the test report(s).

4.2.2 Schedule and sequence of tests. When specified (see 6.2), specific schedule and sequence of component tests are required. Advance planning and provisions shall be made to ensure that each item of the schedule and sequence shall not have a detrimental effect or preclude the performance of a subsequent test/examination on any test sample.

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4.2.3 Advance notice of schedule. Appropriate advance notice of the test schedule, location(s), or changes shall be provided to the Government, contractor, designated test site/facility, and regulatory agency personnel to allow participation or oversight by each.

4.2.4 Certificates of conformance (CoC). When specified (see 6.2), CoC's shall be:

a. Provided by the contractor and in contractor format, except when specified otherwise by this specification, the contract, or a separate scope of work (SOW), with sufficient supporting technical information to ensure adequate evaluation and comment by the Government.

b. Required for delivery to the Government, and may be required for delivery to other activities or agencies to verify safety, HAZMAT, operational, and other requirements

c. Prior to delivery for test or acceptance in order to assure safe operation.

d. Included in all Final Test Reports and, when specified, Conformance Inspection (CI).

4.2.5 Test fluid(s). The test fluid for FAT shall be F24 except where otherwise specified.

4.2.5.1 Recycling and disposal. The use, reuse, and disposal of the required test fluid, lubricants, and other expendable items shall be in compliance with all applicable US statutory, regulatory, and local requirements for the proper storage and disposal of hazardous wastes and waste water discharges. Economic use and recycling for contractor in-process and other testing is encouraged to reduce costs to the Government.

4.2.5.2 Alternate test fluid. Unless specified otherwise, water may be used as the designated military fuels and ASTM test fluids (see 3.4.2) for the following tests when conducted above 33°F.

a. Component testing where exposure to or compatibility with fuels/lubricants is not required; e.g. metallic fittings and connectors.

b. Hose and hose assembly tests in which the distance exceeds one length of hose.

c. As a test fluid media (product) for any CI tests and product demonstrations that are not FAT test requirements.

4.2.6 Prohibitions. The following are specifically prohibited unless specified otherwise within section 4.

a. Lot sampling, acceptable quality limits (AQL), or similar inspection and testing schemes/methods.

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b. Modeling and simulation for all tests and inspections

4.3 Design and construction verification. To verify conformance to 3.3, hose and hose assemblies shall be examined to determine that the hose type is as specified. Nonconformance to 3.3 shall constitute failure.

4.3.1 Hose dimensions.

4.3.1.1 Type A, B, and D hose size. To verify conformance to 3.3.1.1, hose and hose assemblies shall be examined to determine that the hose size is as specified. Nonconformance to table I shall constitute failure.

4.3.1.2 Type C hose size and weight. To verify conformance to 3.3.1.2, hose and hose assemblies shall be examined to determine that the hose size is as specified. A 20-ft section of hose shall be weighed on a scale that measures in not greater than 0.25-lb increments. The weight in pounds per foot shall be calculated. Nonconformance to table II shall constitute failure.

4.3.2 Length. To verify conformance to 3.3.2, hose and hose assemblies shall be measured to determine that the hose length is as specified. Nonconformance to 3.3.2 shall constitute failure.

4.3.3 Tube and cover. To verify conformance to 3.3.3, hose and hose assemblies shall be examined to determine that the hose tube and cover are as specified. Nonconformance to 3.3.3 shall constitute failure.

4.3.4 Static bond. To verify conformance to 3.3.4, hose and hose assemblies shall be examined for presence of a continuous embedded electrical conductor. Electrical resistance between ends of hose, including fittings for hose assemblies, shall be measured during and after subjection to proof test pressure (see 4.3.7.1). Nonconformance to 3.3.4 shall constitute failure of this test.

4.3.5 End fittings. To verify conformance to 3.3.5, hose assemblies shall be examined to determine that the class, style, material, and size of the end fittings are as specified, and that they are securely attached to the hose. Hose assemblies shall also be examined to ensure the end fittings are provided with dust plugs or caps as specified. Nonconformance to 3.3.5 shall constitute failure.

4.3.5.1 Pull strength (assemblies with re-attachable fittings). To verify conformance to 3.3.5.3.1, two re-attachable fittings shall be installed on an appropriately sized hose, approximately 6 ft. in length. The hose assembly shall be hung vertically, by attachment only to the fitting, and used to lift a dead weight of 450 lb not less than 1 in. off the surface and back to it. The test shall be repeated for a total of 10 cycles. On completion of the lifts, the fitting-to-hose interface shall be examined for hose slippage and separation. Hose slippage or separation shall constitute failure of this test. Upon successful completion of the pull test, the hose

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assembly shall be subjected to the proof pressure test, in accordance with 4.3.7.1, for not less than 5 minutes. Any leakage, deformation or separation of the fitting shall constitute failure of this test.

4.3.6 Physical and chemical requirements of bulk hose. To verify conformance to 3.3.6, the tests of 4.3.6.1 through 4.3.6.9 shall be conducted.

4.3.6.1 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation of the cover and tube of the hose shall be determined in accordance with ASTM D412. Nonconformance to table III shall constitute failure of this test.

4.3.6.2 Tensile strength, ultimate elongation, and volume change after fuel immersion. After fuel immersion, the tensile strength, ultimate elongation, and volume change of the cover and tube of the hose shall be determined in accordance with ASTM D471. The immersion time shall be 70 hours. The immersion fuel shall be reference fuel B as specified in ASTM D471. The immediate deteriorated properties method shall be used in determining changes in tensile strength and ultimate elongation. Nonconformance to table III shall constitute failure of this test.

4.3.6.2.1 Tensile strength and ultimate elongation after water immersion (type C hose only). Tensile strength and ultimate elongation of the cover and tube of type C hose shall be determined in accordance with ASTM D471 after immersion in water for 336 hours (14 days). This test shall be repeated after 1008 hours (42 days) of immersion. Nonconformance to table III shall constitute failure of this test.

4.3.6.3 Adhesion.

4.3.6.3.1 Dry adhesion. The original (dry) adhesion between the hose tube and reinforcement shall be determined in accordance with the machine method of ASTM D413, using test specimens. This test shall be repeated to determine the dry adhesion between the hose cover and reinforcement. Nonconformance to table III shall constitute failure of this test.

4.3.6.3.2 Adhesion after immersion in fuel (type A, B, and D hose). The adhesion between the hose tube and reinforcement shall be determined in accordance with the machine method of ASTM D413, using test specimens that have been immersed in reference fuel B as specified in ASTM D471 for a period of 70 hours. The adhesion shall be determined within 15 minutes after removal of the specimens from the fluid. This test shall be repeated to determine the adhesion between the hose cover and reinforcement after immersion in fuel. Nonconformance to table III shall constitute failure of this test.

4.3.6.3.3 Adhesion after fill test (type C hose only). A 12-in. section of hose shall be stoppered and filled with reference fuel B as specified in ASTM D471. The section of hose with the test fluid shall be maintained at a temperature of 160 ±2°F for not less than 46 hours. At the conclusion of the 46-hour cycle, cool the hose to 71 ±9°F and stabilize the temperature. Within 15 minutes after removal of the test fluid, specimens shall be cut from the hose and tested in

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accordance with ASTM D413, machine method. Nonconformance to table III shall constitute failure of this test.

4.3.6.4 Low temperature flexibility.

4.3.6.4.1 Low temperature flexibility (type A, B, and C hose only). A test for low temperature flexibility (for types A and B) and torsional rotation (for type C) shall be conducted in accordance with the routine inspection and acceptance procedures specified in ASTM D1053, except a gaseous media shall be used. The test temperature shall be –25 ±2°F. The exposure time shall be 168 hours. Nonconformance to table III shall constitute failure of this test.

4.3.6.4.2 Brittleness (type D hose only). Prepare test samples of type D hose tube and cover in accordance with ASTM D2137, test method B. Condition the samples in reference fuel B as specified in ASTM D471, for 96 hours. Remove the samples from the fuel and air-dry them for 48 hours. When dry, oven-age the samples for 24 hours at 130 ±2°F, then test the samples in accordance with ASTM D746. Nonconformance to table III shall constitute failure of this test.

4.3.6.4.3 Modulus of elasticity (type D hose only). Using test samples of type D hose tube and cover, determine the modulus of elasticity at –60 ±2°F in accordance with ASTM D790, Type II, and procedure B. The samples shall have been conditioned in reference fuel B as specified in ASTM D471 for 96 hours, then air-dried and oven-aged as specified for brittleness. Nonconformance to table III shall constitute failure of this test.

4.3.6.5 Resistance to aging. Hose samples selected shall be subjected to accelerated aging in accordance with ASTM D573. The samples shall be conditioned in an oven for 96 hours at a temperature of 160 ±2°F. The tensile strength and ultimate elongation shall then be determined on the aged samples. Nonconformance to table III shall constitute failure of this test.

4.3.6.6 Existent gum. The hose shall be tested in accordance with ASTM D381. A hose not less than 14 in. long shall be stoppered at one end with a 2 in. plug and secured with a clamp. The hose shall be filled with Reference Fuel D of ASTM D471 and similarly secured at the opposite end. A sample of the neat fuel shall be retained for later use. The hose shall be tested for a period of seven days by agitating the fuel by means of movement every 24 hour period. The agitating process is performed by moving the hose from a vertical to a horizontal position at a rate of 2 cycles per minute for 5 minutes at each 24 hour period. At the end of the seven day storage period, the fuel shall be agitated in the same manner as above for 5 minutes and then immediately removed from the hose.

Two aliquots of fuel from the hose shall be tested. The existent/unwashed gum content of the fuel shall be determined in accordance with ASTM D381 (air/steam jet vaporizing medium) and an evaporation time of 45 minutes. The residue obtained from the existent/unwashed gum determination, shall then be used to determine the heptane-washed gum content in accordance with ASTM D381. The retained neat fuel sample shall be tested with the same methods. The existent/unwashed gum content from the neat fuel shall be subtracted from

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the existent/unwashed gum content from the fuel used in the soak test. The heptane-washed gum content from the neat fuel shall be subtracted from the heptane-washed gum content from the fuel used in the soak test. Nonconformance to table III shall constitute failure of this test.

4.3.6.7 Crush resistance (type B and D hose). Crush resistance shall be determined by centering a 12-in. length of hose between parallel plates, 3 in. wide, in such a way that a 3-in. length of hose is being compressed. The plates shall be brought together at a rate of 2 in. per minute until the applicable crush resistance load specified in table VIII is reached. The distance between the plates shall be determined to an accuracy of 0.01 in. The results shall be expressed as a percentage of the original hose outside diameter. The static bond (see 4.3.4) shall be measured at maximum crushing force. The load shall be released and the minimum outside diameter of the hose, at the center of the compressed area, shall be measured with the same accuracy. The results shall be expressed as a percentage of the original outside diameter. Nonconformance to 3.3.4 or table III shall constitute failure of this test.

TABLE VIII. Crush resistance and bend radius for types B and D hose.

Hose Nominal I.D. (in)	Crush resistance (lb)	Bend radius (in.)
0.50	200	2.75
0.75	200	3.50
1.00	200	4.00
1.25	200	5.00
1.50	225	6.00
2.00	250	8.00
2.50	325	10.00
3.00	325	12.00
4.00	325	16.00
6.00	325	24.00

4.3.6.8 Bend radius (type B and D hose). A 20-ft section of hose shall be placed in a fixture such that the hose is bent through an angle of 90 degrees at the applicable radius specified in table VIII. The hose shall then be subjected to the appropriate proof pressure for the hose size. Nonconformance to table III shall constitute failure of this test.

4.3.6.9 Ozone resistance. Samples of the cover shall be prepared for testing in accordance with ASTM D1149, method B, procedure B1, static strain. After conditioning for 24 hours in an ozone-free atmosphere, the mounted samples shall be exposed to an atmosphere containing 50 ±10 parts per hundred million (pphm) of ozone, for 336 hours at 104 ±2°F. On completion of the test, the samples shall be examined for visible cracking under a 7X lens. Nonconformance to table III shall constitute failure of this test.

4.3.6.10 Abrasion resistance (type C hose only). The cover of type C hose shall be tested for abrasion resistance in accordance with ASTM D3389. Cut three samples from the

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hose and perform the abrasion appraisal on the cover. The appraisal shall be performed using abrasion wheel H-22 with neither additional weight nor counterweight applied (see 6.2).

4.3.7 Requirements of hose assemblies.

4.3.7.1 Proof pressure. To verify conformance to 3.3.7.1, each length of hose or hose assembly shall be subjected to the hydrostatic tests in accordance with ASTM D380 for elongation, twist, and proof pressure. The test pressure shall be as specified in 3.3.7.1 and shall be maintained for not less than 30 seconds. Nonconformance to 3.3.7.1 shall constitute failure of this test.

4.3.7.2 Burst pressure. To verify conformance to 3.3.7.2, the hose shall be subjected to the straight bursting test in accordance with ASTM D380. Nonconformance to 3.3.7.2 shall constitute failure of this test.

4.4 Interface requirements. To verify conformance to 3.4, CoC's shall be provided to certify compliance to each requirement listed and, in addition, written observations and photographic evidence of compliance shall be included in the FAT Final Test Report. Absence of the CoC, failure of CoC to certify all requirements of 3.4, and absence of the observations and photographic evidence of compliance shall constitute failure.

4.5 Ownership and support requirements. To verify conformance to 3.5, CoC's shall be provided to certify compliance to each requirement listed and, in addition, written observations and photographic evidence of compliance shall be included in the FAT Final Test Report. Absence of the CoC, failure of CoC to certify all requirements of 3.5, and absence of the observations and photographic evidence of compliance shall constitute failure.

4.5.1 CBRN survivability. Conformance to 3.5.5 is a "design to" criteria that is not attainable with current commercial procedures, materials, and production processes, therefore technical and performance testing/evaluation for this criteria shall not be performed. The FAT Final Report shall identify this known shortfall, and list each component that cannot be decontaminated with current procedures.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose and hose assemblies are military unique and intended to be used for the transfer of liquid hydrocarbon fuels in a battlefield environment. The vagaries of the battlefield require petroleum equipment specification to exceed commercially available products in the range of performance. The expected operating terrain and climate are unique and beyond the design of similar commercial items.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Classification (type, size, class, end fitting style, end fitting material, and length) of hose/hose assembly required (see 1.2 and 6.6).

c. When first article inspection is required, time frame for submission of first article test report, and number of units required (see 3.1 and 4.1.1).

d. When static wire is not required (see 3.3.4).

e. When dust plugs or caps are not required (see 3.3.5).

f. Cover color (see 3.5.3). Marking requirements; when contract number is required, when PIN is required (see 3.5.4).

g. When conformance inspection is required (see 4.1.2).

h. When a schedule and sequence of tests is required (see 4.2.2).

i. When a certificate of conformance is required (see 4.2.4).

j. When an abrasion resistance test for type C hose is not required (see 4.3.6.10).

k. Packaging required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a first article production unit(s). The first article should consist of one or more units. The contracting officer should include specific instructions in acquisitions documents regarding arrangements for examinations, approval of the first article examination results, and disposition of the first article(s). Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products who wish

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to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Noncorrosive device. Appropriate hose band specified in SAE J1508 can be used as guidance for noncorrosive devices (see 3.3.5).

6.5 Definitions.

6.5.1 Collapsible hose. Hose or hose in assemblies that do not have integral stiffening and may be collapsed to ribbon form.

6.5.2 Non-collapsible hose. Hose or hose in assemblies that have a spiral or helix of stiffening material added between the cover and tube to provide resistance to collapse.

6.5.3 Recovered materials. Recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

6.5.4 Tube. The inner layer of material in a hose, or hose in an assembly, that is in contact with the product being carried by the hose.

6.5.5 Reinforcement. Synthetic or natural material in fiber, cord, or mat used to provide additional tensile strength.

6.5.6 Cover. The outer layer of material on a hose, or hose in an assembly, that is in contact with the atmosphere.

6.6 NATO fuel designations. The NATO fuel designations F-34, F-35, F-44 and F-54 are defined in STANAG 1135, "Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations" (see 3.4.2).

6.7 MOPP IV gear. MOPP IV protection consists of a two-piece protective overgarment, protective mask with hood, overboots and rubber gloves with cotton liners. The overgarment is closed, and hood is pulled down and adjusted (Field manual FM-3-100) (see 3.5.2).

6.8 Subject term (key word) listing.

Cam-locking
Collapsing
Hoseline
Low temperature
Static bond
Torsional rotation

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Transfer

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

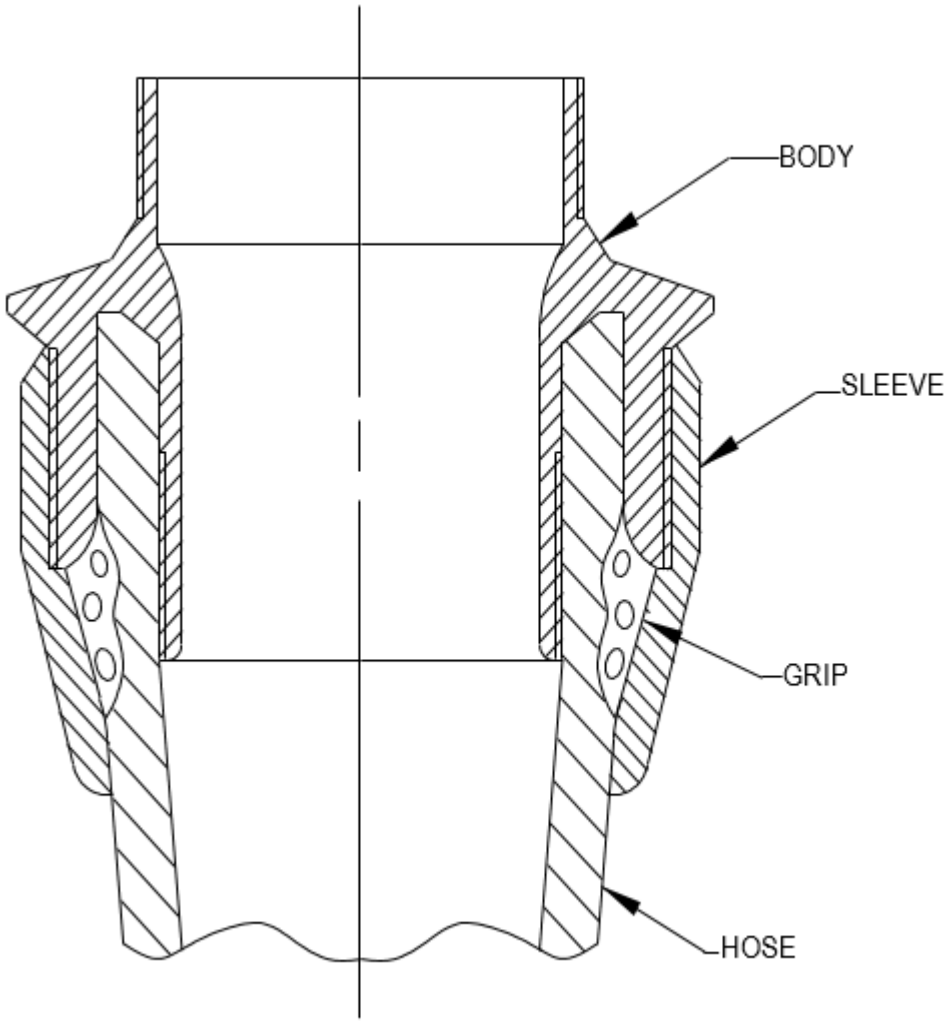


FIGURE 1. Compression type reattachable coupling - For guidance only.

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Custodians:
Army - AT
Navy - SH
Air Force - 184

Preparing activity:
Army - AT

(Project 4720-2017-019)

Review activities:
Air Force – 68, 71
DLA – CC

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